markdown

Handover for Testers — Smart Property Management System

Version 2.0
Author: Samukelo Jili (Lead Developer)
Audience: ENEL2SEH1 project team
Purpose: Step-by-step guide to manually test the SPMS app using VS Code

---

1. GET THE PROJECT FILES

You'll receive one of these from the lead developer:
- A **GitHub link** (recommended), or
- A **`.zip` file** via Google Drive / OneDrive

If GitHub
1. Open the GitHub link in your browser
2. Click the green **"Code"** button → **"Download ZIP"**
3. Extract the zip somewhere sensible:

C:\Users<YourName>\Desktop\SmartPropertyManagementSystem\

text
Do not put it inside OneDrive/Dropbox/Google Drive — file-lock errors will bre
ak the app.

If zip
1. Download the zip
2. Extract to the same path above

You should end up with a folder called `SmartPropertyManagementSystem` containing `main.py`, `ui/`, `Services/`, `Documentation/`, and the rest.

---

2. OPEN THE PROJECT IN VS CODE

1. Launch **VS Code**
2. **File → Open Folder…** (or `Ctrl+K` then `Ctrl+O`)
3. Navigate to `C:\Users\<YourName>\Desktop\SmartPropertyManagementSystem`
4. Click **"Select Folder"**

You should now see the file tree on the left with folders: `ui/`, `Services/`, `Controllers/`, `Repositories/`, `Models/`, `Database/`, `Documentation/`, `Tests/`.

If VS Code asks **"Do you trust the authors of the files in this folder?"** → click **"Yes, I trust the authors"**.

---

3. INSTALL THE REQUIRED VS CODE EXTENSION

You only need **one** extension.

1. Press **`Ctrl+Shift+X`** to open the Extensions panel
2. In the search box, type: **`Python`**
3. Find the one **by Microsoft** (it has a blue checkmark badge and millions of installs)
4. Click **Install**
5. Wait ~30 seconds. VS Code will auto-install **Pylance** as well — you don't need to do anything.

That's all. No other extensions are needed for testing.

---

4. OPEN THE INTEGRATED TERMINAL

All testing commands run in VS Code's built-in terminal. To open it:

- Press **`Ctrl + \`** (Ctrl + backtick key, the one under `Esc`)
- Or **View → Terminal**
- Or **Terminal → New Terminal** from the top menu

A panel opens at the bottom of VS Code. It's already set to the project folder — you don't need to `cd` anywhere.

Verify you're in the right place. Type this and press Enter:

```powershell
pwd
```

It should print `C:\Users\<YourName>\Desktop\SmartPropertyManagementSystem`.

If it doesn't, close the terminal and reopen it. If it still doesn't, close VS Code and re-open the project via **File** → **Open Folder**.

---

5. INSTALL PYTHON (one-time, if not already)

In the VS Code terminal, type:

```
powershell
python --version
```

- **If it prints `Python 3.12.x` or higher** → skip to Step 6.
- **If it says "python is not recognized"** → install Python:
  1. Go to https://www.python.org/downloads/
  2. Download Python 3.12 (or newer)
  3. **On the installer, tick "Add Python to PATH" BEFORE clicking Install** ← critical
  4. After install, **close VS Code completely** and reopen it
  5. Reopen the terminal (`Ctrl + \``) and runpython --version` again

---

6. INSTALL THE REQUIRED PYTHON PACKAGES (one-time)

In the VS Code terminal, run this single command:

```
powershell
python -m pip install PySide6 SQLAlchemy bcrypt reportlab
```

Wait ~60 seconds. You should see `Successfully installed ...` at the end.

Verify it worked:

```
powershell
python -c "import PySide6, sqlalchemy, bcrypt, reportlab; print('OK')"
```

Should print `OK`.

---

7. RUN THE APPLICATION

Still in the VS Code terminal, from the project root:

```
powershell
python main.py
```

The app window opens within ~5 seconds. Leave the terminal open — it shows errors if anything goes wrong.

Test accounts

Role	Email	Password
Administrator	`admin@spm.local`	`admin123`
Property Manager	`manager@spm.local`	`manager123`
Owner (John)	`owner@spm.local`	`owner123`
Owner (Sarah)	`sarah@spm.local`	`sarah123`
Tenant (Jane)	`tenant@spm.local`	`tenant123`
Tenant (Thabo)	`thabo@spm.local`	`thabo123`
Tenant (Naledi)	`naledi@spm.local`	`naledi123`
Inactive (Peter)	`peter@spm.local`	`peter123`

Emails are case-insensitive — `ADMIN@SPM.LOCAL` works the same as `admin@spm.local`. Passwords are case-sensitive.

---

8. YOUR ASSIGNED TEST SUITE

Each team member tests **one role's journey**. Your assignment:

Suite	Tester	Focus
A	(fill in)	Authentication & Access
B	(fill in)	Admin CRUD Operations
C	(fill in)	Role-Based Experiences
D	(fill in)	Notifications, Reports, Dashboard

Full test scenarios for your suite are in:

```
text
Documentation/Testing_Guide.md
```

Open it in VS Code — in the left sidebar, expand `Documentation/` → click `Testing_Guide.md`.

You'll see step-by-step scenarios with an ID (A-01, A-02, etc.), steps, and expected outcome for each. Work through them **in order**, and note any scenario where the app doesn't match the expected outcome.

What each suite covers

Suite A — Authentication & Access

- Can correct credentials log in?
- Are wrong passwords rejected with a visible error?
- Does the app block users trying to log in as the wrong role?
- Is the inactive account rejected?
- Does the "Show/Hide" password toggle work?
- Does the Enter key submit the form?

Suite B — Admin CRUD Operations

- Can admin create, edit, archive Properties, Users, Leases, Payments, Maintenance?
- Are required-field validations enforced?
- Are unique constraints (duplicate email/ID) enforced?
- Are status transitions correct (e.g. property becomes Occupied after lease creation)?

Suite C — Role-Based Experiences

- Login as **tenant** → do they see cards on Properties (not a table)?
- Login as **owner** → do they see only their owned properties?
- Do sidebar items differ between roles (e.g. Users only for admin)?
- Are read-only badges shown where expected?
- Are write buttons hidden for read-only roles?

Suite D — Notifications, Reports, Dashboard

- Does the notification bell show the correct unread count?
- Does clicking a notification mark it read?
- Do PDF reports generate and open?
- Do dashboard stat cards match the actual data in the system?

---

9. HOW TO LOG A BUG

Create a shared Google Doc (send the link to the whole team and to the lead developer) and paste this header:

```
text
| Bug ID | Tester | Date | Module | Severity | Steps to reproduce | Expected | Act
ual | Console output |
|--------|--------|------|--------|----------|--------------------|----------|----
----|----------------|
```

Fill one row per bug you find. Example row:

```
text
| B-001 | Siyavuya | 25/09 | Login | High | Type admin@spm.local + wrong password
→ Sign In | Red error message shows | No message appears | (none) |
```

Severity guide:

- **Critical** — app crashes, data loss, security hole
- **High** — feature doesn't work at all
- **Medium** — feature works but has a logic or visual bug
- **Low** — cosmetic (typo, misaligned text)

How to capture the console output

If the app prints a red traceback in the VS Code terminal when something breaks:

1. Click inside the terminal panel
2. Select the red error text with your mouse
3. Right-click → **Copy**
4. Paste into the "Console output" column of your bug row

---

10. TROUBLESHOOTING (VS Code Edition)

"No module named PySide6" (or SQLAlchemy, bcrypt, reportlab)

Re-run the install command in the VS Code terminal:

```
powershell
python -m pip install PySide6 SQLAlchemy bcrypt reportlab
```

The window opens then closes instantly

Run `python main.py` from the VS Code terminal (not from the file explorer). The terminal keeps the error visible after the app closes — screenshot it and send to the lead developer.

The app is blank or frozen

1. Close the app window
2. In the VS Code terminal:

   ```
   powershell

   Get-ChildItem -Recurse -Filter "__pycache__" | Remove-Item -Recurse -Force
   ```
3. Run `python main.py` again

"cannot open file" or "database is locked"

Close the app and reopen it. If it persists, delete `Database/smartproperty.db` and run:

```
powershell
python -m Database.seed_rich_data
```

This recreates the rich test data.

VS Code terminal shows the wrong folder

Close the terminal (`Ctrl + ``) and reopen it. If it's still wrong, close VS Code and use **File** → **Open Folder** to reopen the project.

"I can't find Document X"

All documents are in the `Documentation/` folder. Open the VS Code sidebar and expand `Documentation/`. Key files:

File	Purpose
`Handover_For_Testers.md`	This file
`Testing_Guide.md`	Full test scenarios per suite
`Software_Project_Master_Document.docx`	Complete project spec
`ENEL2SEH1 - Project Phases.docx`	Submission draft
`reports/` (folder)	Generated PDFs land here

---

11. TIMELINE

Milestone	Date
Receive project + this doc	(fill in)
Complete your assigned suite	(fill in)
Submit your bug log + sign-off	(fill in)
Lead dev fixes bugs	+ 3 days

Milestone	Date
Re-test fixes (optional)	+ 5 days
Phase 6 (Testing) submission	(fill in)

---

12. SIGN-OFF

When you complete your suite, fill in your row in the shared bug doc:

Tester	Suite	Date completed	Bugs found	Status
	A			Pass / Fail
	B			Pass / Fail
	C			Pass / Fail
	D			Pass / Fail

Phase 6 is complete when all four rows are filled.

---

13. WHO TO CONTACT

Issue type	Contact
Bug found	Post in the shared bug doc + tag the lead developer
Blocked for more than 15 min	Message the lead developer directly
Can't run the app at all	Send a screenshot of the VS Code terminal to the lead developer

Lead developer: Samukelo Jili
Team leader: Sphumelele Ngidi

Don't spend more than **15 minutes stuck** on the same problem. Screenshot, message, move on to another scenario.

```
text

---

## Quick Reference — What the Team Actually Does

| Step | Action | Where |
|------|--------|-------|
| 1 | Download the project | GitHub or zip link |
| 2 | Extract to Desktop | Windows Explorer |
| 3 | Open VS Code → File → Open Folder | VS Code |
| 4 | Install "Python" extension | VS Code Extensions (`Ctrl+Shift+X`) |
| 5 | Open terminal | `Ctrl + \`` in VS Code |
| 6 | Install packages | `python -m pip install PySide6 SQLAlchemy bcrypt reportla
b` |
| 7 | Run app | `python main.py` |
| 8 | Read `Documentation/Testing_Guide.md` | VS Code sidebar |
| 9 | Work through their assigned suite | The running app |
| 10 | Log bugs in shared Google Doc | Browser |
| 11 | Sign off | Shared doc |

That's it. No terminal juggling, no path confusion — everything happens inside VS
Code.

---
```